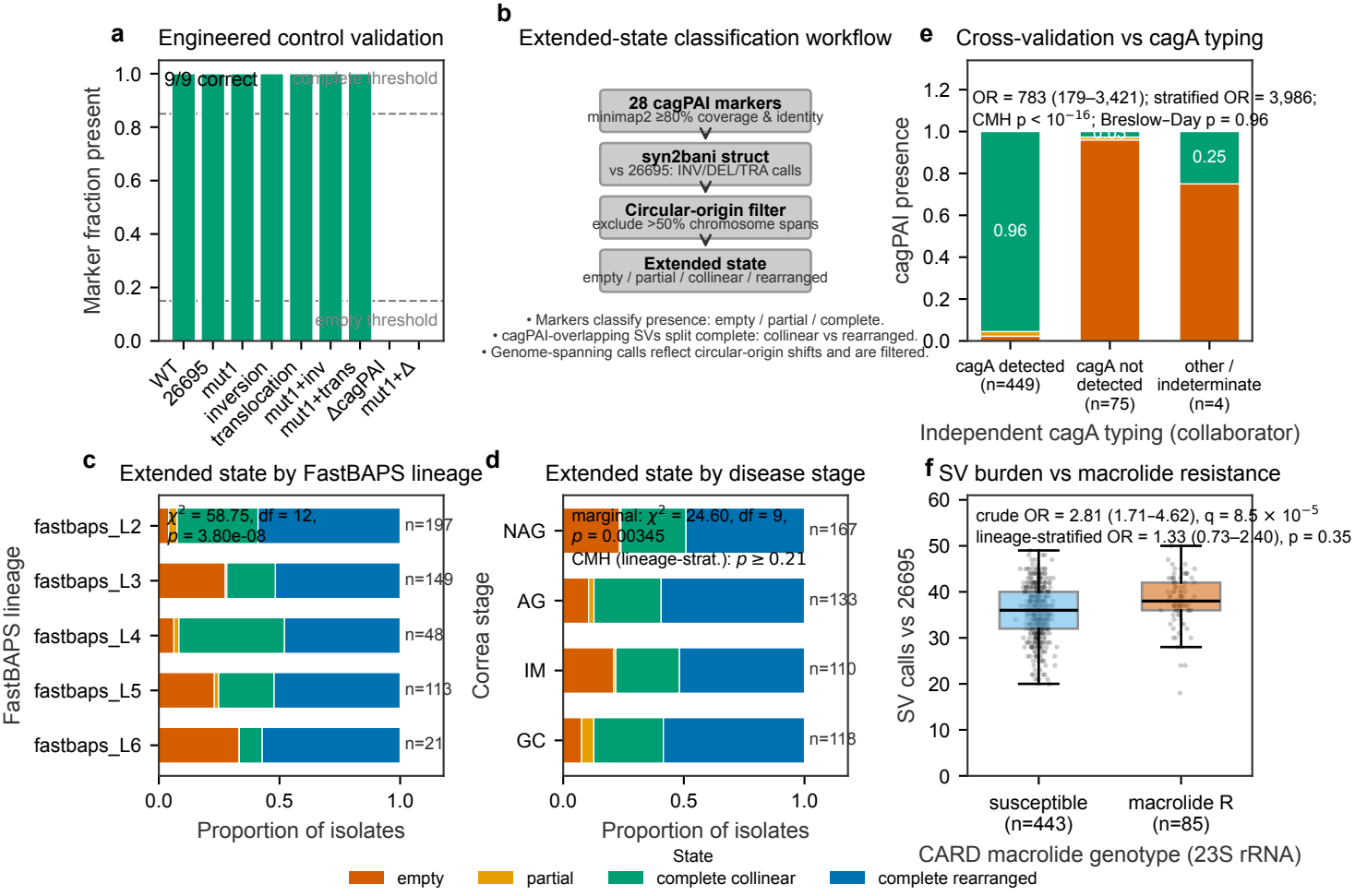

a
Engineered control validation
9/9 correct
complete threshold
empty threshold
Marker fraction present
1.0
0.8
0.6
0.4
0.2
0.0
WT
26695
mut1
inversion
translocation
mut1+inv
mut1+trans
ΔcagPAI
mut1+Δ
b
Extended-state classification workflow
28 cagPAI markers
minimap2 ≥80% coverage & identity
syn2bani struct
vs 26695: INV/DEL/TRA calls
Circular-origin filter
exclude >50% chromosome spans
Extended state
empty / partial / collinear / rearranged
• Markers classify presence: empty / partial / complete.
• cagPAI-overlapping SVs split complete: collinear vs rearranged.
• Genome-spanning calls reflect circular-origin shifts and are filtered.
e
Cross-validation vs cagA typing
OR = 783 (179–3,421); stratified OR = 3,986;
CMH p < 10−16; Breslow–Day p = 0.96
cagPAI presence
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.96
0.25
cagA detected
(n=449)
cagA not
detected
(n=75)
other /
indeterminate
(n=4)
Independent cagA typing (collaborator)
c
Extended state by FastBAPS lineage
χ² = 58.75, df = 12,
p = 3.80e-08
FastBAPS lineage
fastbaps_L2
fastbaps_L3
fastbaps_L4
fastbaps_L5
fastbaps_L6
n=197
n=149
n=48
n=113
n=21
0.0
0.5
1.0
Proportion of isolates
d
Extended state by disease stage
marginal: χ² = 24.60, df = 9,
p = 0.00345
CMH (lineage-strat.): p ≥ 0.21
Correa stage
NAG
AG
IM
GC
n=167
n=133
n=110
n=118
0.0
0.5
1.0
Proportion of isolates
State
empty
partial
complete collinear
complete rearranged
f
SV burden vs macrolide resistance
crude OR = 2.81 (1.71–4.62), q = 8.5 × 10−5
lineage-stratified OR = 1.33 (0.73–2.40), p = 0.35
SV calls vs 26695
60
50
40
30
20
10
0
susceptible
(n=443)
macrolide R
(n=85)
CARD macrolide genotype (23S rRNA)